

SRv6 uSID

Roadmap

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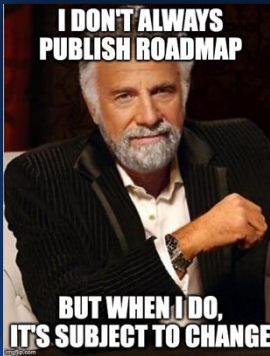


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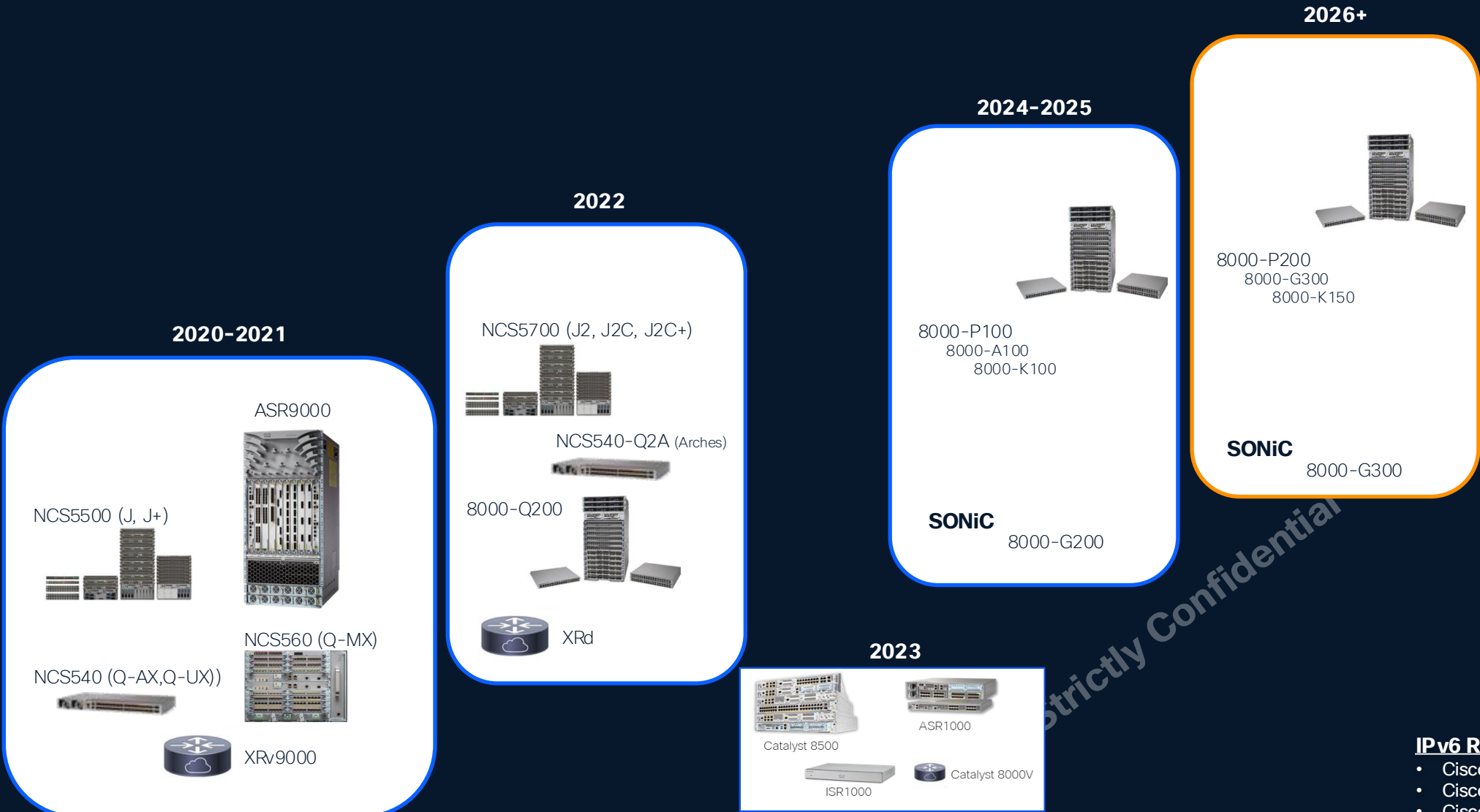


Many of the products and features described herein remain in varying stages of development and will be offered on a when-and-if-available basis.

This roadmap is subject to change at the sole discretion of Cisco, and Cisco will have no liability for delay in the delivery or failure to deliver any of the products or features set forth in this document.



SRv6 uSID Platform Support



- IPv6 Routing**
- Cisco 12000
 - Cisco NCS 6000
 - Cisco CRS-1
 - Cisco 8000 (Q100)
 - ASR900 (XE)

XR: SRv6 Roadmap

Available Now

Roadmap (27.1.1 +)

- SRv6 Transport (ISIS)
- TiLFA, uLoop (ISIS)
- Flex-Algo (ISIS)
- L3 Services: VPNv4, VPNv6, IPv4/IPv6 Inet
- L2 Services: EVPN-VPWS ELINE
- SRv6 OAM (Ping/Trace, SID Verification)
- SRv6/MPLS Interworking Gateway
- SRv6/MPLS Dual-Connected PE
- SRv6 uSID (CSID-NEXT)
- SRv6 PCC/PCE (PCEPv6) w/ ODN
- SR-PM Link, Endpoint Delay/Loss
- L2 Services: EVPN ELAN
- SRv6-TE Explicit Path
- Path Tracing
- UPA
- Explicit Anycast End.DT46 using W-LIB
- Deterministic Demand Matrix
- SRv6 PMTU support
- SRTE gRPC API
- Per-Flow Manual Steering
- EVPN-IRB: Distributed Anycast Gateway
- IP Measurement (IPM) w/ 3L
- SRv6 Multicast
- VxLAN <-> SRv6 DCI GW
- DSDN: Auto-BW
- SRv6 w/ IPSec
- L3-DAG
- PW Headend
- Mo6: Transport Interworking
- DSDN: Circuit-Style SRv6TE
- DSDN: Congestion Mitigation

CY2018-21

CY22-23

CY24-26

CY2027+

NCS5500



NCS540



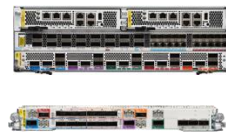
ASR9000



NCS560



ASR9903
+ 5th Gen (LightSpeed+)



NCS540-L (Felidae, Darwin)



NCS5700 (J2, J2C)



NCS540-Q2A (Arches)



8000-Q200
(Gibraltar)



8000-P100/K100/A100
(Palladium)
(Krypton)
(Argon)



8000-P200/K150

SONiC

8000-O : G200

8000-O : G300



SONiC: SRv6 Roadmap

Shipping

1. Scale-out

- G200 (Release: 202511.1.1.0)
- IPV6 forwarding with BGP control plane
- **SRv6 uSID transit behavior / SRv6 uN shift and forward**
- Host based packet spraying support to avoid ECMP hash collisions
- Telemetry/PFC/ECN

2HCY26

2. Scale-across

- Extend support to G200x/G300
- Extend shift-forward for scale across
- **SRv6 uSID support (encap / decap) – Required for AI edge.**
- Packet trimming for congestion recovery
- G-series first; P200 for scale
- Telemetry/PFC/ECN

1HCY27

3. Lean DCI

- **P200: Static SRv6 TE**
- Controller-driven policy programming
- Backup tunnel on primary-path failure
- NHG/FRR behavior and interop evidence

Host / NIC owns

Path spraying, flow control, retransmission, path retirement, and congestion/failure reaction.

Switch must do

Static SRv6/uSID shift-forward, packet trimming, ECN/MMU behavior, telemetry, counters, and deterministic forwarding.

XE: SRv6 Roadmap

Available Now

Roadmap

- SRv6 Transport (ISIS)
- SRv6 uSID (CSID-NEXT)
- SRv6 over IPsec tunnel mode
- L3 Services: VPNv4, VPNv6, IPv4/IPv6 Inet
- L3 Services: Explicit SRv6 L3VPN
- SRv6 OAM (Ping/Trace, SID Verification)
- SR-PM Link Delay Measurement

- TiLFA, uLoop (ISIS)
- SR-PM Liveness of SRv6TE policy
- Per-Flow Manual Steering
- Static Route for IPv4/IPv6
- SRv6/MPLS Dual-Connected PE
- SRv6 PCC w/ ODN+AS
- SD-WAN Overlay + SR Underlay (2-box)
- SRv6-TE OAM
- SRv6 PMTU support

- Flex Algo (ISIS)
- Flex Algo with TiLFA
- SRv6 eBGP
- SRv6-TE PCC (PCEP)
- MVPN Ingress Replication
- CNC support for IOS XE
- L2 Services: EVPN-VPWS (DX2)
- CNC support for G2 Routers

- L2 Services: EVPN ELAN (DT2U/DT2M)
- Inline NAT for SRv6 Traffic
- ISIS CMAC authentication
- UPA
- SRv6 transport with SGT
- App QoE
- Oper-models Support

CY23-24

CY24-25

CY26

CY2027+



Catalyst 8500,
8500L
Catalyst 8300
Catalyst 8200



ASR1000 Fixed
HX-Series

ASR1006-X
ASR1009
(with RP3 +
ESP100-X or RP3
+ ESP200-X)



ISR1000



Catalyst 8000V



Cisco 8000 Series
Secure Routers

NXOS: SRv6 Roadmap

Committed

1. DCI GW

- **Encapsulation:** SRv6 uSID (Supported formats: F3216, F4816)
- **DCI Handoff:** VXLAN EVPN ↔ SRv6 uSID VPN gateway, Border-device for routing/forwarding between fabrics and core
- **OAM / Verification:** uSID-aware SRv6 OAM, Path validation and fault isolation

Committed CY27

Simplify **Data Center Interconnectivity** and enhance network efficiency by seamlessly bridging VXLAN EVPN fabrics with scalable, high-performance SRv6 uSID technology

Planned

2. SRv6 AI Backend

- **Source-Routing:** Host driven traffic engineering
- **Path enumeration:** Explicit SID list definition for path mapping
- **Faster convergence:** Custom flow steering at the compute edge

Unlock maximum AI cluster performance with an **SRv6-enabled backend and NIC(host-based overlay)**, providing real-time, deterministic acceleration for every GPU flow.

Future

3. SRv6 in the DC

- **Encapsulation:** SRv6 uSID (Supported formats: F3216, F4816)
- **Native IPv6 Forwarding:** Leverages the existing IPv6 underlay for transport, reducing encapsulation overhead
- **OAM / Verification:** uSID-aware SRv6 OAM, Path validation and fault isolation

Unify and streamline multi-tenant fabrics with **SRv6 EVPN**, a high-performance, IPv6-native architecture built for simplified, programmable service.

How can Cisco help you?

Cisco® Services Portfolio

Resiliency. Simplicity. Time to value.



Cisco Services Portfolio

Flexible tiers and commercial models tailored to your evolving business need

1 Cisco Support

AI-driven and expert-led support, designed to help you realize the value from your IT investments.

Basic

- ✓ Device-level issue resolution with 60-min response time
- ✓ Unified asset, service coverage, and risk reports
- ✓ Self-serve troubleshooting and case management

Standard plus Basic

- ✓ Centralized triage of solution issues with 30-min response time
- ✓ Prioritized lifecycle, case, and service coverage insights
- ✓ Curated security, field notice, and security hardening assessment insights
- ✓ Automated configuration best practice insights
- ✓ Cisco U. personalized learning paths

Signature plus Standard

- ✓ Expert-led support with 15-min response time
- ✓ Restoration Service Level Agreement
- ✓ Tailored predictive disruption analysis and prevention
- ✓ Personalized asset management
- ✓ Expert-led systematic root cause analysis and continuous optimization
- ✓ Automated security hardening and configuration recommendations
- ✓ Cisco U. certification training

2 Cisco Professional Services

Expert guidance to accelerate transformation, reduce risk, optimize operations, and deliver business outcomes.



Subscription

Ongoing services through flexible subscription model



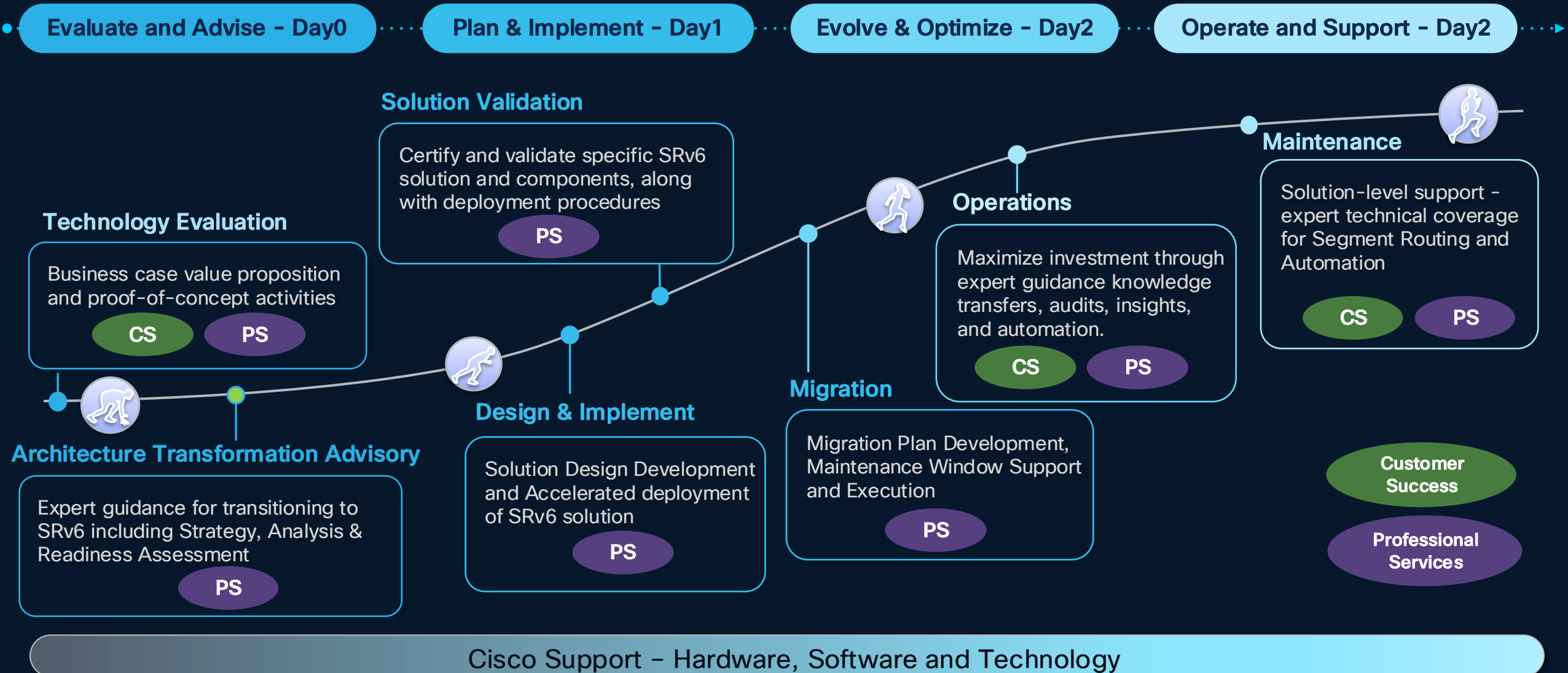
Transactional

Project-based services

Enterprise Agreement

Cisco Services for SRv6

Engagement model across deployment stages: Customer Success, Cisco Support and Professional Services



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